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UNIT III | Chapter 6

Respiration

Chapter Outline

6.1 Respiratory function
6.2 Respiratory organs in various organisms
6.3 Mechanism of breathing
6.4 Exchange of gases
6.5 Transport of gases
6.6 Regulation of respiration
6.7 Problems in oxygen transport
6.8 Disorders of respiratory system
6.9 Effects of smoking



Exercise increases the rate and depth of breathing and supplies extra oxygen to the muscles and removes more CO_2 from the tissues.

Learning Objectives:

- *Learns to describe the gross structure of the human gaseous exchange system*
- *Observes and draws the tissues and organs associated with the respiratory system*
- *Understands the process of gaseous exchange and transport of respiratory gases*
- *Knows the problems associated with oxygen transport*
- *Gains knowledge on the ill–effects of smoking.*

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We inhale and exhale air. Why is breathing so important for life? What happens when we breathe? Why energy is required for the body to perform various life processes? Where does the energy come from? We eat food for energy. Though the above raised questions look disconnected, we should know that the process of breathing is connected to the process of release of energy from food. Oxygen is utilized by the organisms to breakdown the biomolecules like glucose and to derive energy. During this breakdown carbondioxide, which is a harmful gas is also released. It is very obvious that oxygen has to be provided to cells continuously and the CO_2 to be released immediately by the cells. So the need of a respiratory system is essential for life.